

Primary VPT Chain: ProducerUA -> ProZorro -> Retail

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely I(1)-like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

standardized_type=butter

retailer=novus

promo_variant=promo_controlled

frequency=daily

Primary chain is strictly ProducerUA -> ProZorro -> Retail. Retail is estimated separately for silpo, novus, and silpo_novus. Combined rule: daily median of available silpo and novus standardized_type prices.

Table: PreTests

series	adf_level_p	kpss_level_p	adf_diff1_p	kpss_diff1_p	adf_diff2_p	kpss_diff2_p	integration_class	stability_flag	standardized_type	retailer	promo_variant	cointegration_p	pair_n_obs
producer	0.08225840900720094	0.01	2.6818026212664985e-12	0.1	1.3006957963949979e-20	0.1	I(1)	1.0	butter	novus	promo_controlled	nan	nan
prozorro	0.05640422694264767	0.01	3.518946761497874e-22	0.1	1.9269184182015793e-13	0.1	I(1)	1.0	butter	novus	promo_controlled	nan	nan
retail	nan	nan	nan	nan	nan	nan	ambiguous	0.0	butter	novus	promo_controlled	nan	nan
producer_to_prozorro	nan	nan	nan	nan	nan	nan	pair	nan	butter	novus	promo_controlled	0.46279135586174147	220.0
prozorro_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	butter	novus	promo_controlled	nan	17.0
producer_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	butter	novus	promo_controlled	nan	16.0

Table: ModelCoefficients

standardized_type	retailer	promo_variant	frequency	link	model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry	short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	notes	primary_chain_doc
butter	novus	promo_controlled	daily	producer_to_prozorro	NARDL	prozorro	producer	218	-0.2127764229425364	0.27405320585302084	-0.8083252331134949	1.2523745815882856e-26	0.787381486847105	0.20112635504343956		nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...	

Table: ResidualDiagnostics

model_family	link	y_series	x_series	ljungbox_p	arch_p	jb_p	unreliable_flag
NARDL	producer_to_prozorro	prozorro	producer	0.30437404399912776	0.5287391650637225	4.2203813853019655e-84	0

Table: SeriesUsed

date	producer	prozorro	retail	standardized_type	retailer	promo_variant	combined_rule
2021-02-01	31.78268739915295	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-02	31.74153855497394	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-03	31.71205395008976	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-04	31.69228954461795	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-05	31.68030129867603	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-06	31.67414517238153	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-07	31.67187712585198	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-08	31.6715531192049	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-09	31.71501589339797	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-10	31.82710410052748	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-11	31.98036756741884	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-12	32.14735612089748	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-13	32.30061958778885	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-14	32.41270779491835	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-15	32.45617056911143	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-16	32.45341900628176	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-17	32.44522030312711	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-18	32.43165843764893	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-19	32.41281738784873	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-20	32.38878113172797	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-21	32.35963364728814	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-22	32.32545891253073	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-23	31.91148084818262	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-24	30.90284056637734	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-25	29.53949783222626	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-26	28.06141241084077	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-27	26.70854406733228	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-02-28	25.72085256681216	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-03-01	25.33829767439181	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-03-02	25.36591783597624	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-03-03	25.44026511447978	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-03-04	25.54856970052784	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-03-05	25.67806178474581	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-03-06	25.81597155775909	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-03-07	25.94952921019307	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-03-08	26.06596493267315	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-03-09	26.18214222460376	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-03-10	26.31503339825227	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-03-11	26.45203186353847	nan	nan	butter	novus	promo_controlled	daily_median_of_available...
2021-03-12	26.58053103038216	nan	nan	butter	novus	promo_controlled	daily_median_of_available...

Table: LagProfile

	pair	lag	corr
producer_to_prozorro	1	0.07873337531291687	
producer_to_prozorro	2	0.08283043430904553	
producer_to_prozorro	3	0.08251800718380085	
producer_to_prozorro	4	0.08422490303011267	
producer_to_prozorro	5	0.0907848031216401	
producer_to_prozorro	6	0.08185920607839353	
producer_to_prozorro	7	0.08099913022176998	
producer_to_prozorro	8	0.08418185337829556	
producer_to_prozorro	9	0.08011971167494186	
producer_to_prozorro	10	0.08071177659888301	
producer_to_prozorro	11	0.0855520813606708	
producer_to_prozorro	12	0.08891786146062777	
producer_to_prozorro	13	0.0906935444584109	
producer_to_prozorro	14	0.09336004953876562	
producer_to_prozorro	15	0.09806722582073225	
producer_to_prozorro	16	0.10411102969445078	
producer_to_prozorro	17	0.11083781027515911	
producer_to_prozorro	18	0.11554030327110705	
producer_to_prozorro	19	0.11683243707057434	
producer_to_prozorro	20	0.11514579002515733	
producer_to_prozorro	21	0.11293989092313386	
producer_to_prozorro	22	0.11348584116046659	
producer_to_prozorro	23	0.11614446866624657	
producer_to_prozorro	24	0.1193793796780393	
producer_to_prozorro	25	0.12118388938915296	
producer_to_prozorro	26	0.11984324455041497	
producer_to_prozorro	27	0.11226973352035723	
producer_to_prozorro	28	0.1007561552537856	
producer_to_prozorro	29	0.09278063269691687	
producer_to_prozorro	30	0.09120495228803738	
prozorro_to_retail	1	-0.45878211653703	
prozorro_to_retail	2	0.19735414941647134	
prozorro_to_retail	3	-0.09043719970230159	
prozorro_to_retail	4	-0.5435987792757687	
prozorro_to_retail	5	0.1892047653885967	
prozorro_to_retail	6	-0.19203204120456288	
prozorro_to_retail	7	-0.005128401453782567	
prozorro_to_retail	8	-0.3601954141406435	
prozorro_to_retail	9	-0.32544457516371006	
prozorro_to_retail	10	-0.3172181443665881	

Table: ModelEligibility

standardized_type	retailer	promo_variant	frequency	integration_producer	integration_prozorro	integration_retail	n_obs_producer	n_obs_prozorro	n_obs_retail	n_obs_pair_prod_prozorro	n_obs_pair_prozorro_retail	any_i2	eligibility_note
butter	novus	promo_controlled	daily	I(1)	I(1)	ambiguous	1793	230	19	220	17	0	Low pair overlap in main ...

Table: NARDL_Multipliers

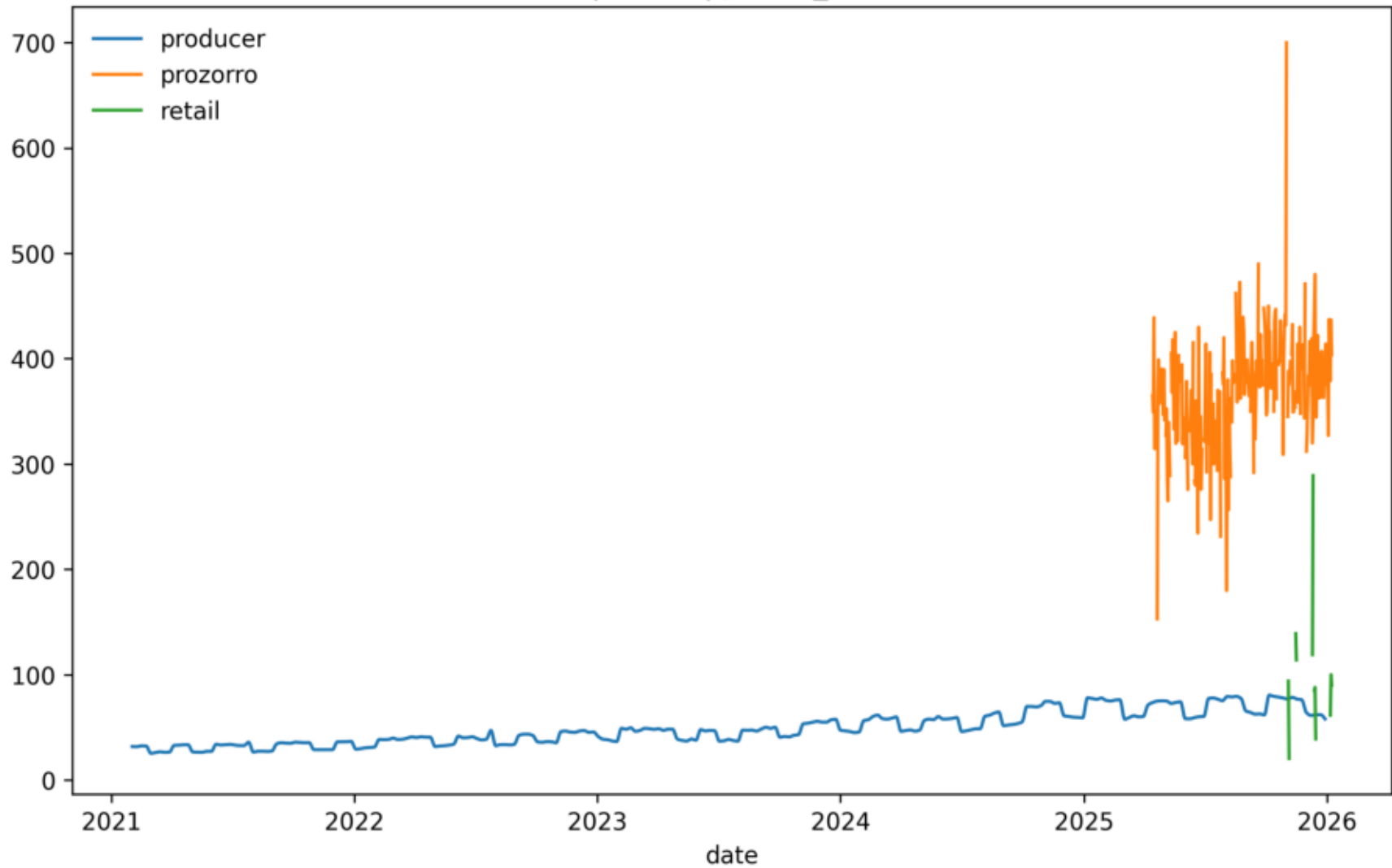
horizon	mult_pos		mult_neg	link
0	-0.5437076425441754	-0.33093121960163896	producer_to_prozorro	
1	-0.48043463639437756	-0.2924196897746266	producer_to_prozorro	
2	-0.48779791951561324	-0.29690140029865086	producer_to_prozorro	
3	-0.4869410305889971	-0.2963798492381926	producer_to_prozorro	
4	-0.4870407495079179	-0.29644054381166113	producer_to_prozorro	
5	-0.4870291448967695	-0.29643348058903957	producer_to_prozorro	
6	-0.48703049536267273	-0.2964343025589613	producer_to_prozorro	
7	-0.4870303382046144	-0.2964342069036792	producer_to_prozorro	
8	-0.4870303564936028	-0.2964342180353919	producer_to_prozorro	
9	-0.48703035436525444	-0.2964342167399587	producer_to_prozorro	
10	-0.4870303546129372	-0.2964342168907124	producer_to_prozorro	
11	-0.4870303545841136	-0.29643421687316873	producer_to_prozorro	
12	-0.48703035458746785	-0.2964342168752103	producer_to_prozorro	
13	-0.4870303545870775	-0.29643421687497273	producer_to_prozorro	
14	-0.48703035458712296	-0.2964342168750004	producer_to_prozorro	
15	-0.48703035458711763	-0.29643421687499716	producer_to_prozorro	
16	-0.4870303545871183	-0.29643421687499755	producer_to_prozorro	
17	-0.4870303545871182	-0.2964342168749975	producer_to_prozorro	
18	-0.4870303545871182	-0.2964342168749975	producer_to_prozorro	
19	-0.4870303545871182	-0.2964342168749975	producer_to_prozorro	
20	-0.4870303545871182	-0.2964342168749975	producer_to_prozorro	

Table: VECM_IRF

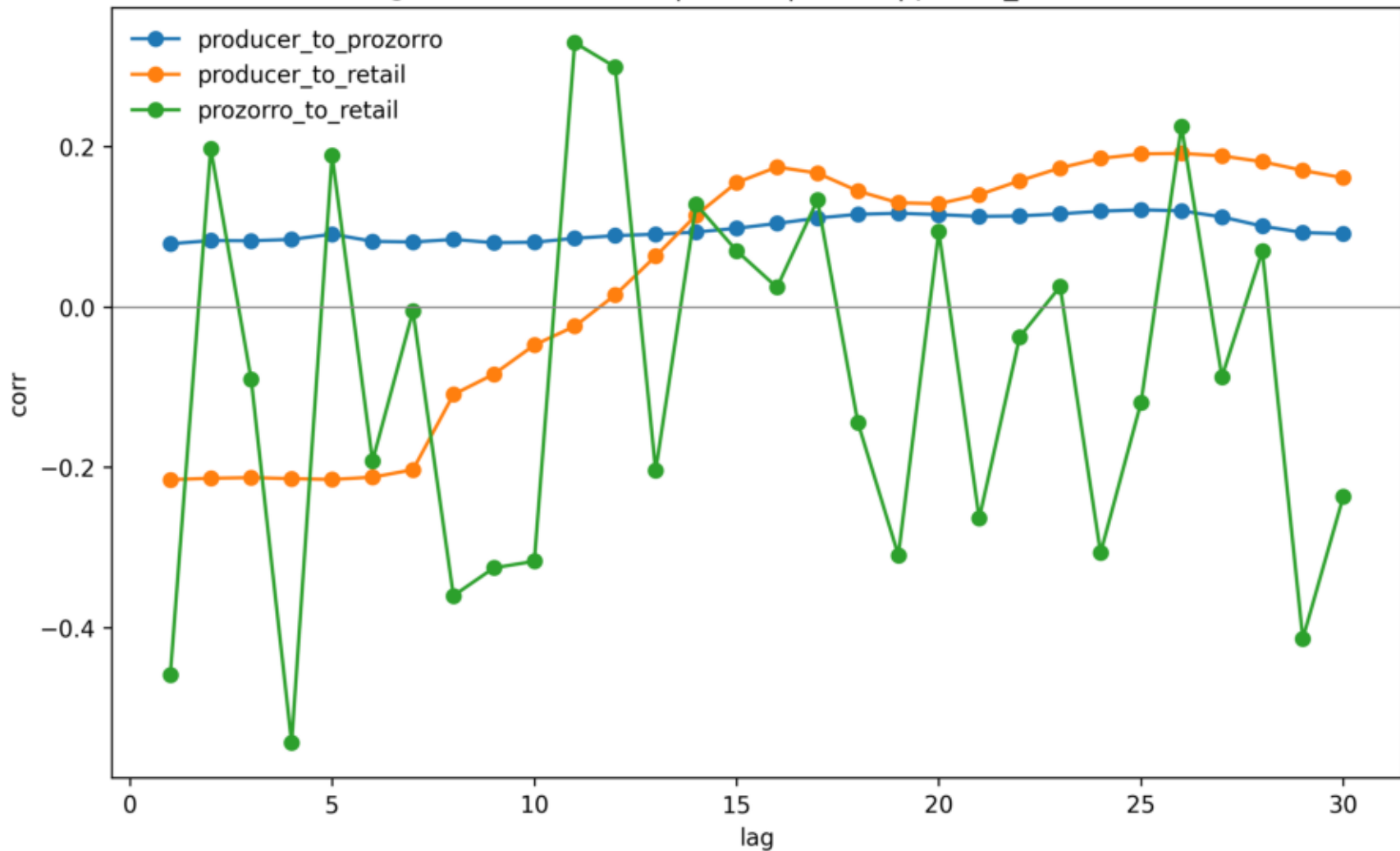
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time_series_promo_controlled.png

butter | novus | promo_controlled



Lag-Correlation Profile | butter | novus | promo_controlled



NARDL Dynamic Multipliers | butter | novus | promo_controlled

